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$$\begin{aligned} -\nabla \cdot (A \nabla u) &= f & \text{on } \Omega \\ u &= u_0 & \text{on } \Gamma \end{aligned}$$

$A(\vec{x}) \in \mathbb{R}^{2 \times 2}$ is symmetric & PD $\forall \vec{x}$, $f(\vec{x})$ is a given distributed force, & $u_0(\vec{x})$ is the source on boundary Γ

(a) Derive weak form & the energy functional that is minimized by the solution of u .

To get weak form integrate strong form w/ a test function $\hat{u}(\vec{x})$:
where $\hat{u}(\vec{x}) \in H_0^1$, s.t. $\hat{u}(\vec{x}) = 0$ on Γ

$$\int_{\Omega} -\nabla \cdot (A \nabla u) \hat{u}(\vec{x}) \, d\vec{x} = \int_{\Omega} f(\vec{x}) \hat{u}(\vec{x}) \, d\vec{x}$$

w/ Green's Identity:

$$\int_{\Omega} A \nabla u \cdot \nabla \hat{u}(\vec{x}) \, d\vec{x} - \int_{\Gamma} \hat{u}(\vec{x}) A \nabla u \cdot \vec{n} \, ds = \int_{\Omega} f(\vec{x}) \hat{u}(\vec{x}) \, d\vec{x}$$

$\nearrow 0 \text{ on } \Gamma$

$$\boxed{\int_{\Omega} A \nabla u \cdot \nabla \hat{u} \, d\vec{x} = \int_{\Omega} f(\vec{x}) \hat{u} \, d\vec{x}} \quad \text{weak form of PDE w/BC:}$$

$u = u_0 \text{ on } \Gamma$

The first variation of the energy functional $\delta_u(\Pi(u, \hat{u})) = 0$ is equivalent to the weak form of the PDE:

$$\begin{aligned} \delta_u \Pi(u, \hat{u}) &= \int_{\Omega} A \nabla u \cdot \nabla \hat{u} \, d\vec{x} - \int_{\Omega} f(\vec{x}) \hat{u} \, d\vec{x} = 0 \\ &= \int_{\Omega} A \nabla u \cdot \nabla \hat{u} - f \hat{u} \, d\vec{x} = 0 \quad \forall \hat{u} \in H_0^1 \end{aligned}$$

Since $\delta_u \Pi(u, \hat{u}) = \frac{d}{d\varepsilon} \Pi(u + \varepsilon \hat{u}) \Big|_{\varepsilon=0}$, notice that to get a

$\int_{\Omega} \nabla u \cdot \nabla \hat{u} \, d\vec{x}$ term, the energy functional $\Pi(u)$ must have a squared ∇u term.

3a cont) consider $\pi_1 = \int_{\Omega} \nabla u \cdot \nabla u dx$. $\delta_u \pi_1(u, \hat{u}) = \frac{d}{d\epsilon} (\pi(u + \epsilon \hat{u})) \Big|_{\epsilon=0}$ 23

$$\delta_u \pi_1(u, \hat{u}) = \frac{d}{d\epsilon} \left[\int_{\Omega} \nabla(u + \epsilon \hat{u}) \cdot \nabla(u + \epsilon \hat{u}) dx \right] \Big|_{\epsilon=0} = 2 \int_{\Omega} \nabla(u + \epsilon \hat{u}) \cdot \nabla \hat{u} dx \Big|_{\epsilon=0} \\ = 2 \int_{\Omega} \nabla u \cdot \nabla \hat{u} dx$$

for a $\int_{\Omega} \hat{u} dx$ term in $\delta_u \pi_1(u, \hat{u})$, the energy functional must have a u term. consider $\pi_2 = \int_{\Omega} u dx$; $\delta_u \pi_2(u, \hat{u}) = \frac{d}{d\epsilon} \left[\int_{\Omega} (u + \epsilon \hat{u}) dx \right] \Big|_{\epsilon=0} = \int_{\Omega} \hat{u} dx$

So, the energy functional is:

$$\pi(u) = \frac{1}{2} \int_{\Omega} A \nabla u \cdot \nabla u dx - \int_{\Omega} f(x) u dx$$

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$$A_2 = \begin{pmatrix} 1 & -5 \\ -5 & 100 \end{pmatrix}$$

To find the eigenvalues/vectors:

$$(A_2 - \lambda I) \vec{v} = 0; |(A_2 - \lambda I)| = 0$$

$$\begin{vmatrix} 1-\lambda & -5 \\ -5 & 100-\lambda \end{vmatrix} = (1-\lambda)(100-\lambda) - 25 = 0$$

$$\lambda = \frac{101 \pm \sqrt{9901}}{2} \approx 100.25, 0.75$$

For the eigenvectors:

$$(A_2 - \lambda_1 I) \vec{v}_1 = 0$$

more than 100 orders
of magnitude difference
- highly anisotropic

$$\begin{pmatrix} -99.25 & -5 \\ -5 & -0.25 \end{pmatrix} \vec{v}_1 = \vec{0} \Rightarrow \vec{v}_1 \approx \begin{bmatrix} 1 \\ -19.85 \end{bmatrix}$$

$$(A_2 - \lambda_2 I) \vec{v}_2 = 0$$

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$$\begin{pmatrix} .25 & -5 \\ -5 & 99.25 \end{pmatrix} \vec{v}_2 = 0 ; \vec{v}_2 \approx \begin{pmatrix} 19.85 \\ 1 \end{pmatrix}$$

Note that for the 2 eigenvectors, the eigenvalue magnitude is vastly different, indicating more than a 100x conductivity along $\vec{v}_1$ than $\vec{v}_2$.

For visual clarity, we can check the conductivity magnitude along the x & y axes. Note that these axes do not correspond with the eigenvectors.

$$\vec{x}^T A \vec{x} = (1 \ 0) \begin{pmatrix} 1 & -5 \\ -5 & 100 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1$$

$$\vec{y}^T A \vec{y} = (0 \ 1) \begin{pmatrix} 1 & -5 \\ -5 & 100 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 100$$

So, the tensor conducts 100x more along the y direction than the x direction, further displaying anisotropy.